continued growth of the locomotive in America may be justified.

AUTOMATIC STOKERS FOR LOCOMOTIVES.

The necessity for an automatic stoker for the large locomotive, either passenger or freight, is quite generally recognized, and the requirements have been met in a very satisfactory manner by at least one form of stoker. It is strange, therefore, that a stoker which has been upon the market for several years, and which works so successfully, does not become regularly adopted by the railroads, which seem to need some such appliance to assist the fireman in his arduous work. The testimony as to the success of this device is the most convincing of any relating to locomotive improvements (containing such radical changes in methods of operation) which have been introduced in many years. The reports of the laboratory tests at Purdue show it to have given a very satisfactory performance. The testimony of motive power men who have had quite a number of stokers in use is entirely favorable to the device, and the opinion of the superintendent of our largest locomotive works is that the stoker is the coming device.

PISTON VALVES.

The flat slide valve is almost always provided with balancing strips, but, on account of imperfect lubrication, its excessive friction makes it difficult for the engine driver to change the cut-off, and the want of sufficient rigidity in the ordinary design has led to the use of a very heavy valve gear on modern locomotives. The piston valve was introduced for the purpose of securing a balanced valve, thus reducing valve friction. It has been designed in various forms, and much experimental work has been done with packing rings of different shapes. Although many objections have been made to the piston valve on account of broken packing rings and leakage, yet it has been gradually developed and successfully applied, so that it may be regarded as well established practice in America.

The use of the piston valve has modified materially the design of the cylinder and saddle, as well as the front frames, and in such designs there is opportunity to make a stronger construction of both cylinder and frame than with the flat valve. The packing rings used by the Baldwin Locomotive Works are rectangular and nearly square. The construction used by the American Locomotive Company for packing rings includes a bushing, with L-shaped rings on each side of it, and a large number of locomotives are now running with this form of packing ring.

The principal objection which has been urged against the piston valve is the loss due to leakage, and it was supposed that this was much greater than with flat valves. The subject was carefully investigated by a committee of the Master Mechanics' Association, and the following figures are taken from a report presented at the convention of that association in June, 1904. The amount of leakage per hour was measured with each type of valve, one series with valves at rest and another with valves in motion. The best results obtained from piston valves show a leakage of 268 lbs. per hour, and from flat valves, 348 lbs. The worst cases of leakage were 2,880 lbs. with piston valves and 2,610 lbs. with flat valves.

Incidentally, it was found that the principal cause of leakage was poor fitting of the rings, and the conclusion was reached that if equal attention is given to each kind of valve, the piston valve would show less leakage. Flat valves are free to lift from the seat, and thus relieve the cylinders from water and undue compression, but piston valves cannot lift, and provision must be made for the rapid removal of water from the cylinder; therefore it is essential that relief valves and well-designed by-pass valves be used in connection with piston valves.

The question of the lubrication of piston valves while drifting is one which is still in the experimental stage, but it is also one to which proper attention must be given, in order to make the use of such valves entirely satisfactory.

VALVE GEAR.

The use of the Walschaert valve motion on the Baltimore & Ohio, Mallet articulated locomotive, and on the Pennsylvania de Glehn four-cylinder compound, which are now on exhibition at St. Louis, has again brought the merits of this gear to the attention of American designers. These two locomotives represent extreme conditions as to the speeds for which they are intended. The one is for slow speed on heavy grades, the other for high-speed passenger work. This would indicate that the Walschaert gear is well adapted to any kind of service—freight or passenger. It is fortunate that a well-designed gear of this type will soon be seen in operation in the United States, and its performance on these locomotives will be watched with interest.

This gear has been used for many years on the State Railways of Belgium, and it is also used extensively in Germany and France. In the latter country it has been given preference over all others for the high-speed balanced compounds, which have made such remarkable records, and, on this account, it is found on the French locomotive which was built for the Pennsylvania.

The chief difference between the Walschaert and the Stephenson motions is the constant lead with the former when the valve travel is changed. This is due to the fact that at the end of the stroke the cross-head alone determines the position of the valve, and as the cross-head always has the same position at the end of the stroke the valve will also have a definite location, and the travel may be decreased but the lead remains constant. For high-speed locomotives, of the ordinary simple two-cylinder type, the constant lead may not be regarded as desirable, as early cut-offs are then used, and it is necessary to have greater pre-admission, when the cut-off is so short, in order to permit the steam to enter the cylinder without excessive wire-drawing. With the four-cylinder balanced compound, which is the most promising type for high-speed work, the cut-off need not be short. The record of the indicator cards taken from a locomotive of this kind on the Northern of France, as given in Mr. Sauvage's paper, shows that at 77 miles per hour the cut-off was 45 per cent. in the high-pressure cylinder and 67 per cent. in the low-pressure cylinder. The Joy valve gear has a constant lead, and it is used quite generally in England in connection with inside cylinders, and it has not been found objectionable on account of this peculiarity. As far as the distribution of steam is concerned, the Walschaert valve motion will produce results as good as, if not better than, the Stephenson shifting link, and it has some mechanical advantages which should recommend it as well adapted to modern locomotive practice in the United States.

A valve gear outside of the frames is conveniently inspected and repaired, while one inside of the frames is certainly in an awkward position for either operation. With inside cylinders and crank axles there is little room for eccentrics and links, and if all this be removed it allows ample length for main pin bearings, and it is then possible to have an inside bearing for the crank axle. The Walschaert gear, as ordinarily designed, is not symmetrical in a vertical plane, and there is a tendency to lateral bending and unequal wear when so constructed. In the engine referred to, Mr. de Glehn has taken special care to avoid these objections, and his design shows well-balanced wearing surfaces of ample proportions, which should be quite durable.

The point to which the writer wishes to call particular attention is the great contrast in the weight of the moving parts and the size of the bearings when this Walschaert outside gear is compared with similar parts of a Stephenson link motion driven by eccentrics. A prominent superintendent of motive power, who has made a special study of indicator cards, and who has given much attention to valve gears, in a recent discussion on "Modern Tendencies in Locomotive Design in America," made this statement:

"I consider that the increased complication and weight of the valve motion is an exceedingly serious matter in giving distorted steam distribution, due to the destructive effect of the valve motion in causing wear and tear."

The reports on "Weights of Detail Parts of Locomotives"* give the weights of parts of the Stephenson valve gear for large locomotives, as follows, in pounds: Eccentric 212, eccentric strap 225, eccentric rod 125, link 148, rocker arm 248, transmission bar 128, valve rod 66, valve yoke 90, valve 211. These figures indicate that the Stephenson valve gear, including the eccentrics and straps, as found on modern locomotives, has become a very ponderous affair. Some attention has been given to the valve pattern, in the effort to make it as light as possible, but the same care has not been taken with the moving details connected with it, and which easily become a disturbing factor at high speeds if made too heavy.

The principal load which comes on the eccentrics and straps, causing them to heat, is not the friction of the valve, but it is that due to the inertia of the reciprocating parts of the valve gear, the motion of which is reversed twice for every revolution. If the rocker arm is included, the weight, as found above, of the moving parts from valve to eccentric strap for one cylinder is 1,052 lbs., and at high speeds the energy of this moving mass must impose a heavy load on the eccentrics.

The eccentrics and straps are the most difficult details in the locomotive machinery to keep lubricated properly, and it requires constant vigilance to prevent them from heating. When they do heat and cut, and the straps are taken down, their location inside the frames is the most inconvenient possible, and with the increasing weight of the machinery this part of the locomotive repairs has become very laborious and expensive. More attention should be given to the reduction of the weight of the moving parts of the Stephenson valve gear, or some other type should be used. The Walschaert gear, located outsided the frames, is easily accessible, and very convenient for inspection, lubrication and repairs. The main driving bearings are two small pins with bushed bearings, and the contrast with the heavy and cumbersome eccentrics and straps, which are their equivalent in a valve gear system, is very striking. This gear is simple and light throughout, and it has much in it to be recommended for overcoming the objectionable features of the shifting link motion driven by eccentrics.

LOCOMOTIVE FRAMES.

In the design of locomotive frames the bar frame is still the general practice, and, for the pedestal portion, it does not seem feasible to adapt any other form to the gen-

*Proceedings Master Mechanics' Association, 1903, p. 187.